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EDUCATION

Chonnam National University Sep 2023 – Aug 2025
Master's in Chemistry, Specialization: Biochemistry
Gwangju, South Korea

- GPA: 4.38/4.5

University of Antananarivo Mar 2018 – Oct 2021
Bachelor's in Chemistry
Antananarivo, Madagascar

- GPA: 3.62/4

EXPERIENCE

Graduate Research Student Sep 2023 – Aug 2025
Biochemistry Lab, Chonnam National University
Gwangju, South Korea

- Investigated the DNA-binding properties of *E. coli* aspartate ammonia-lyase and its specific interactions with DNA using Fluorescence spectroscopy.
- Proposed a mechanism of the allosteric activation by metal ions of *E. coli* aspartate ammonia-lyase by identifying its allosteric site through AI-guided site-directed mutagenesis and kinetic assays.

Intern, Chemistry Lab, Institut Pasteur of Madagascar – Feb – May 2022
Antananarivo, Madagascar

- Laboratory assistant in water quality control (qualitative analysis, risk assessment) and food analysis (finished product testing, regulatory compliance, toxic pollutant assessment).
- Characterization of pesticide extraction methods using GC/MS and LC/MS.

Volunteer, Y'Dago, Association of Young Volunteers – Oct 2021 – Jun 2022
Antananarivo, Madagascar

- Collaborator on the project Y'Speak, serving as a volunteer English language teacher.
- Advocate for environmental protection.

CONFERENCES & PRESENTATIONS

Korean Society for Molecular and Cellular Biology – Winter Conference 2025 Jan 2025
Pyeongchang, South Korea

Poster presentation: "Metal-Ion Dependency of Aspartase from *Escherichia coli*"

FAOBMB 2025 May 2025
Busan, South Korea

Poster presentation: "Mutational Analysis Unveils the Metal-Ion Binding Site in *Escherichia coli* Aspartase"

SCHOLARSHIPS & AWARDS

Global Korea Scholarship (GKS)

Sep 2022 – Aug 2025

Awarded in 2022 by the Government of South Korea

CERTIFICATIONS, SKILLS & INTERESTS

Certifications:

- **IELTS** (Overall: 8, R: 8.5, L: 8.5, W: 8, S: 7.5) – 20/05/2024
- **DALF C1** (Overall: 72.5/100, R: 18.5, L: 21.5, W: 17.5, S: 15) – 12/02/2026
- **Test of Proficiency in Korean** level 6 – 18/12/2024

Key Skills:

- **Technical Skills:** Cell culture, protein expression and purification, recombinant DNA technology, Polymerase Chain Reaction (PCR), site-directed mutagenesis, enzyme assays, EMSA, GC/MS, HPLC, FPLC, NMR, UV-Vis and fluorescence spectroscopy, water quality analysis (DO, TDS, COD, BOD), volumetric titration, calorimetric analysis, and sample preparation and extraction techniques.
- **Bioinformatics Skills:** protein structural analysis (PyMOL, UCSF Chimera), molecular docking (AlphaFold, Chai-1), multiple sequence alignment (MAFFT, MUSCLE), phylogenetic analysis (IQ-TREE), MestReNova and TopSpin.
- **Computer Skills:** Microsoft Office, LaTeX, Origin (data analysis and figure preparation), Python, R, Adobe Illustrator, Linux (bioinformatics workflows).
- **Languages:** Malagasy (Native), English (Fluent), French (Fluent), Korean (Advanced).

Interests: programming, foreign languages, outdoor activities, and sports.